

The Cybersecurity Career Starter Guide

2026 Edition • by Kevin Zelman

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Before you start

I wrote the first version of this guide years ago for the people I was mentoring. Most of what was in it has changed: the training sites, the certifications, the tools, and even what an entry-level job looks like now that AI is part of the work. So I rewrote it from scratch.

This guide is for you if you're a student, a career changer, or already in IT (help desk, desktop support, sysadmin) and want to move toward security. It won't make you an expert. It will give you a map, a first plan, and a list of free places to practice.

One promise I'll ask you to keep: only practice on systems you own or have written permission to test. Labs, CTFs and your own home lab are where you learn. Everything else is someone else's network.

Why cybersecurity, and what's true in 2026

- **The need is real.** Every organization with data, email or a website needs people who can protect it, and that includes schools, hospitals, nonprofits and small businesses, not just banks and tech companies.
- **The entry point is competitive.** Security job postings often ask for experience. Most people don't start in a "security" title. They start in IT support, networking or systems work and move over. That's not a detour; it's how you learn how things actually break.
- **Security is a team sport.** The best security people understand identity, networks, cloud, endpoints and people. If you're already fixing laptops and resetting passwords, you're closer than you think.
- **You don't need a degree, but you do need proof.** Degrees help. So do certifications. What moves hiring managers most is evidence: a home lab, write-ups, a GitHub, a competition result, or a story about a problem you solved.

Pick a direction (you can change it later)

Job titles vary a lot between organizations. Here are the common paths and what the day-to-day looks like.

- **Security / SOC Analyst (blue team):** watches alerts, investigates suspicious activity, and escalates or responds. A common first security job.
- **Incident Responder:** takes over when something has actually happened: contains it, figures out what occurred, and helps recover.
- **Identity & Access (IAM) Engineer:** runs sign-in, single sign-on, MFA and permissions. Most attacks now start with a stolen login, so this is one of the most important and in-demand areas.
- **Cloud Security Engineer:** secures Microsoft Azure, AWS or Google Cloud environments and Microsoft 365 or Google Workspace tenants.
- **Security Engineer / Architect:** designs and builds the protections (firewalls, endpoint security, email security, logging) and makes them work together.
- **GRC (Governance, Risk & Compliance) Analyst:** policies, risk assessments, audits and frameworks like NIST CSF, CIS Controls and ISO 27001. Great for people who write well and like structure.

- **Penetration Tester / Red Team:** legally attacks systems to find weaknesses before criminals do. Popular, and usually not a first job; most pentesters come from IT, development or the SOC.
- **Digital Forensics & Threat Intelligence:** studies evidence, malware and attacker behavior.
- **Application / Product Security:** works with developers to build software securely. A natural path if you already code.
- **AI Security:** a newer area covering how organizations use AI tools safely and how AI systems themselves can be attacked or misused.

A realistic starting ladder: Help Desk / IT Support → Systems or Network Administrator (or Junior SOC Analyst) → a security specialty. Each step teaches you something the next one needs.

Tip: the NICE Workforce Framework (from NIST) and CyberSeek.org both map security roles to the skills they need. They're useful for deciding what to learn next.

The skills that matter most

You don't need all of these on day one. Aim to be comfortable with the first five, and curious about the rest.

- **Networking fundamentals:** IP addressing, subnets, DNS, DHCP, ports and protocols, how a firewall and a VPN work.
- **Operating systems:** Windows and Linux at the command line. How users, permissions, services and logs work on each.
- **Identity:** Active Directory and Microsoft Entra ID (or Okta and Google), MFA, single sign-on, least privilege.
- **Cloud basics:** what's different when servers, email and files live in Microsoft 365, Azure or AWS.
- **Scripting:** PowerShell for Windows and Microsoft 365, Python for everything else, and a bit of Bash. Start by automating one boring task.
- **Logs and detection:** reading event logs, and using a SIEM or search tool (Microsoft Sentinel, Splunk, Elastic) to find patterns.
- **Communication and writing:** writing a clear ticket, an incident summary or a how-to guide is a security skill. It's also the one most people skip.

Certifications: a 2026 roadmap

Certifications get you past résumé filters and give your studying structure. Pick them by the job you want next, not by what sounds impressive. Exam versions change, so always check the vendor's site for the current exam and price.

Getting started (no experience needed)

- **CompTIA A+:** hardware, operating systems and troubleshooting. Best if you're aiming for help desk.
- **CompTIA Network+:** networking fundamentals. Useful for every path.
- **ISC2 Certified in Cybersecurity (CC):** an entry-level security cert. ISC2 has offered free training and exams for it through its "One Million Certified" program, so check whether that's still available.
- **Google Cybersecurity Certificate** (on Coursera): a guided beginner program with hands-on labs.
- **Microsoft SC-900** (Security, Compliance & Identity Fundamentals) and **AZ-900** (Azure Fundamentals): short, affordable, and a good introduction to the Microsoft world that many employers run on.

Your first "real" security cert

- **CompTIA Security+**: still the most-requested entry-level security certification, and required for many U.S. government and contractor roles.

Specialize (after a year or two of hands-on work)

- **Blue team / SOC**: CompTIA CySA+, Microsoft SC-200 (Security Operations Analyst), or Security Blue Team's BTL1.
- **Identity**: Microsoft SC-300 (Identity and Access Administrator). This is the one I'd pick if you work in a Microsoft shop.
- **Cloud**: Microsoft AZ-104 then AZ-500, or AWS Cloud Practitioner then AWS Security Specialty.
- **Offensive**: CompTIA PenTest+, eJPT, and later OSCP.
- **GRC**: ISACA CISA or CRISC, or ISC2 CGRC.

Later in your career

- **ISC2 CISSP**: requires five years of relevant experience. You can pass the exam earlier and hold "Associate of ISC2" status until you have the experience.
- **ISACA CISM**: for security management and leadership.

Money-saving tips: many Microsoft exams have free "Virtual Training Days" or exam-discount offers, Microsoft Learn's training is free, and student and military discounts are common. Ask your employer about paying for exams; many will.

Free places to learn and practice

Structured learning

- **Microsoft Learn** (learn.microsoft.com): free learning paths and sandboxes for every Microsoft exam.
- **Professor Messer** (prolessormesser.com): free video courses for A+, Network+ and Security+.
- **Cisco Networking Academy** (netacad.com): free intro courses in networking and cybersecurity.
- **TryHackMe**: guided, browser-based rooms. The free tier is plenty to start.
- **Hack The Box Academy**: in-depth modules, with a mix of free and paid content.

Practice and competitions

- **National Cyber League (NCL)**: a student-friendly competition run through Cyber Skyline, with seasons in spring and fall. It's where I competed, and it's the best structured practice I know.
- **picoCTF**: free, year-round challenges from Carnegie Mellon. Great for beginners.
- **OverTheWire** (Bandit first): learn Linux by playing a game.
- **Blue team practice**: Blue Team Labs Online, CyberDefenders and LetsDefend give you real-looking alerts and investigations.
- **CTFtime.org**: a calendar of CTF competitions worldwide.

Stay current

- **CISA** (cisa.gov) alerts and the Known Exploited Vulnerabilities catalogue.
- **Podcasts**: Darknet Diaries (stories) and the SANS Internet Storm Center daily podcast (five minutes of news).
- **Security conference talks** on YouTube (DEF CON, BSides), many of them free.
- Local **BSides** conferences, OWASP chapters and ISSA/ISACA meetups. These are cheap or free and great for meeting people who hire.

Build a home lab (for \$0 to start)

A home lab is where you learn by breaking things, and it gives you stories to tell in interviews.

- **Virtualization:** install VirtualBox, VMware Workstation or Hyper-V (built into Windows Pro).
- **Build a mini company:** a Windows Server domain controller, one Windows client and one Linux machine. Create users and groups, join the client to the domain, and apply a Group Policy.
- **Add the cloud:** create an Azure free account and explore Microsoft Entra ID. The Microsoft 365 Developer Program has tightened who gets a free tenant, so check eligibility; Microsoft Learn sandboxes are a good fallback.
- **Watch it:** send logs to a free SIEM (Elastic, Wazuh or Splunk's free tier) and learn what normal looks like.
- **Attack it (your own lab only):** use Kali Linux or Parrot OS against deliberately vulnerable machines like Metasploitable or the vulnerable VMs from VulnHub.
- **Document everything:** keep notes on what you built, what broke and how you fixed it. Those notes become blog posts, GitHub READMEs and interview answers.

AI and your career in 2026

AI tools are now part of IT and security work. Used well, they make you faster. Used badly, they create new risks.

- **Use AI as a tutor, not a crutch.** Ask it to explain a log entry, a command or a concept, then check the answer against official documentation. If you can't explain it without the AI, you haven't learned it yet.
- **Never paste sensitive data into an AI tool** unless your organization has approved that tool for that data. That includes passwords, customer information, internal documents and logs with personal details.
- **Learn the new risks:** prompt injection, data leakage, over-shared permissions and AI-generated phishing. The OWASP Top 10 for LLM Applications is a good starting point.
- **Expect entry-level work to change.** Routine tasks such as first-pass alert triage, password resets and basic scripting are increasingly automated. The people who stand out understand why something is happening, can judge whether the tool is right, and can explain it to others.

Turning skills into a job

- **Your résumé:** lead with results, not duties. "Reset passwords" becomes "Resolved 40+ weekly account tickets and wrote the guide that cut repeat calls." Put certifications, labs and competition results near the top if your job history is thin.
- **LinkedIn:** a clear headline about the work you want, a short About section in your own voice, and your home-lab and CTF projects in the Projects section.
- **A portfolio:** a free GitHub Pages site or a GitHub profile with your scripts, lab write-ups and notes. (Never publish anything from an employer, and only publish CTF write-ups after the event ends.)
- **Interviews:** practice explaining your troubleshooting out loud: what you checked, in what order, and why. For behavioral questions, tell short stories with a situation, what you did and the result.
- **Networking:** tell people what you're working toward. Some of my best opportunities came from people who had seen my work.
- **Get in the door:** help desk, IT support, NOC and MSP roles are real entry points. So are internships, student-worker jobs and volunteering IT help for a nonprofit.

Your first 90 days

- **Days 1–30:** pick a direction. Start Network+ or Security+ study (Professor Messer plus practice questions). Finish TryHackMe's beginner path. Join a community.
- **Days 31–60:** build the home-lab domain. Write three short lab write-ups. Automate one task with PowerShell or Python. Update LinkedIn and your résumé.
- **Days 61–90:** take your first exam. Enter a CTF (NCL, picoCTF or one on CTFtime). Apply to 10 roles that match where you are now, and ask one person in the field for a 20-minute conversation.

Stay in touch

I run a free mentoring community for people growing in IT and cybersecurity. 1:1 mentoring is available, and workshops and meetings are held semi-regularly.

- **Discord:** discord.gg/WJn4bQV9m9
- **Mentoring page:** kevinzelman.github.io/mentoring.html
- **LinkedIn:** linkedin.com/in/kevin-zelman

Everyone started somewhere. Ask the question.

Links and programs change. Everything here was current as of fall 2026, but check the provider's site before you pay for anything. This guide reflects my personal views, not those of any employer.